

# The Influence of Lid to the Stress of Underfill in Flip-Chip Package

Xu Peng#

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
xu.peng@siat.ac.cn

Cheng Zhong#

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
cheng.zhong1@siat.ac.cn

Chenglong Li

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*School of Mechanical Science and Engineering, Huazhong University of Science and Technology*  
Wuhan, 430074, China  
iclears@163.com

Yulong Li

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
yl.li@siat.ac.cn

Ruoyu Jiang

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
jry@mail.ustc.edu.cn

Gang Li

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
gang.li@siat.ac.cn

Pengli Zhu

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
pengli.zhu@siat.ac.cn

Jibao Lu\*

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
jibao.lu@siat.ac.cn

Rong Sun\*

*Shenzhen Institute of Advanced Electronic Materials, Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
*Shenzhen Institute of Advanced Technology, Chinese Academy of Science*  
Shenzhen 518055, China  
rong.sun@siat.ac.cn

**Abstract**—In flip-chip package, in order to improve the reliability of solder joints, underfill (UF) are usually used. At present, the package used for evaluating UF reliability is lidless in most case, which is different to the actual flip-chip structure with metal lid. Thus, the influence of lid with different structures and material properties to the stress of underfill is a question worthy of study. In this paper, based on finite element analysis (FEA), the influence of CTE, elastic modulus and thickness of lid on package warpage and UF stress are analyzed in detail. We find that the lid will indeed significantly affect the deformation or warpage in package. For lid with different CTE, three different warpage modes are presented. However, it is worth noting that for stress of UF, larger lid CTE will lead to larger UF interface stress, regardless of the warpage in package. The mechanism of deformation and stress in package is explained in detail. In summary, we believe that the stress level of the UF-related interface will be significantly underestimated in lidless structure.

**Keywords**—Underfill (UF), metal lid, flip-chip package, finite element analysis (FEA)

## I. INTRODUCTION

In flip-chip package, the chip and substrate are connected through grid array of solder joints, which may provide higher I/O density and faster transfer rates. In order to improve the reliability of solder joints, underfill (UF) are usually used [1-2]. The cured UF is closely connected to solder joints, chips, substrates, etc., which helps to reduce the mismatch of coefficients of thermal expansion (CTE) inside package, thereby reducing stress on solders. For application, whether a specific UF can be effectively used in specific package needs to be evaluated by detailed reliability tests. However, the actual flip-chip structure is complex, and advanced large-size packages generally need to add a metal lid for improving thermal conduction and mechanical protection [3-6]. The current problem is that the package used for evaluating UF reliability is lidless in most case, which is different to the actual flip-chip structure with metal lid. Thus, the influence of lid with different structures and material properties to the stress of UF is a question worthy of study.

In this paper, based on finite element analysis (FEA), the influence of CTE, elastic modulus and thickness of lid on

package warpage and UF stress are analyzed in detail. We find that the lid will indeed significantly affect the deformation or warpage in package. For lid with different CTE, three different warpage modes are presented. However, it is worth noting that for stress of UF, larger lid CTE will lead to larger UF interface stress, regardless of the warpage in package. The mechanism of deformation and stress in package is explained in detail. In summary, we believe that the stress level of the UF-related interface will be significantly underestimated in lidless structure.

## II. METHOD

### A. FEA Model

This paper studied a typical flip chip package, including lid, lid attach, die, TIM, UF, solder and substrate. Table 1 lists the actual package dimensions. The model in FEA is constructed according to the actual structure of the package, as shown in Fig. 1.

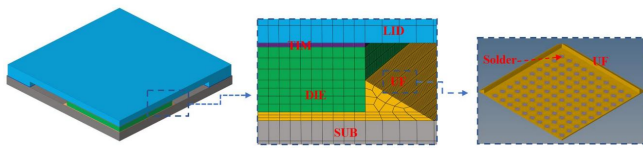


Fig. 1. FEA model.

TABLE I  
GEOMETRIC PARAMETERS

| Part          | Length and Width (mm) | Thickness (mm) |
|---------------|-----------------------|----------------|
| Lid           | 44*44                 | 1.84           |
| Lid attach    | -                     | 0.05           |
| TIM           | 21*21                 | 0.045          |
| Die           | 21*21                 | 0.745          |
| UF and Solder | 21*21                 | 0.1            |
| Substrate     | 45*45                 | 1.38           |

### B. Material Properties

To simplify calculations, the material in the package is linear elastic, as shown in Table 2.

### C. Boundary Conditions

In order to simplify the calculation, we adopt the encapsulated 1/4 structure to study. A cooling load of 165 °C to 25 °C was applied to the simulation model.

## III. ANALYSIS AND DISCUSSION

### A. The Influence of Lid on Warpage

First of all, we study the effect of the lid on warpage of the package. The warpage is mainly determined by the relative value of CTE between lid and substrate. Here, the CTE of substrate is fixed as 14 ppm/°C, and the CTE of lid is considered to be 10, 14, and 18 ppm/°C respectively. The deformation of package is shown in Fig. 2. Clearly, the

deformation trends are quite different. For the lidless package, the chip area was arched upwards, while the edge area of substrate was deformed downwards. When the CTE of lid is 10 ppm/°C, the deformation trend is basically the same as that of the lidless package. However, due to the constraint of lid, the degree of downward deformation at the edge of substrate is reduced. When the CTE of lid is 14 ppm/°C, since the CTE of Lid is consistent with the CTE of substrate, the edge area of the package is nearly flat, but the central area will still be arched upward. Meanwhile, when the CTE of lid is 18 ppm/°C, due to the larger CTE of lid compared to substrate, the edge area of lid trend to shrink more during cooling process. When constrained by the substrate, the entire package will lift up at the edges. At the same time, the middle area of the package will still be arched upwards. In Fig. 2, it clearly shows that the deformation of package is closely related to CTE of lid, and the relative value of CTE between lid and substrate may cause three different deformation modes.

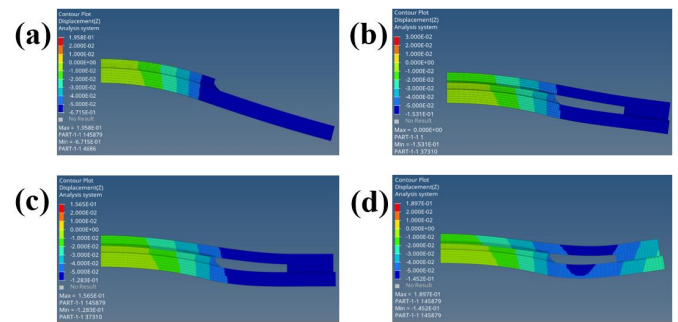


Fig. 2. Package warpage (a) without lid; (b) when the CTE of lid is 10 ppm/°C; (c) when the CTE of lid is 14 ppm/°C; (d) when the CTE of lid is 18 ppm/°C.

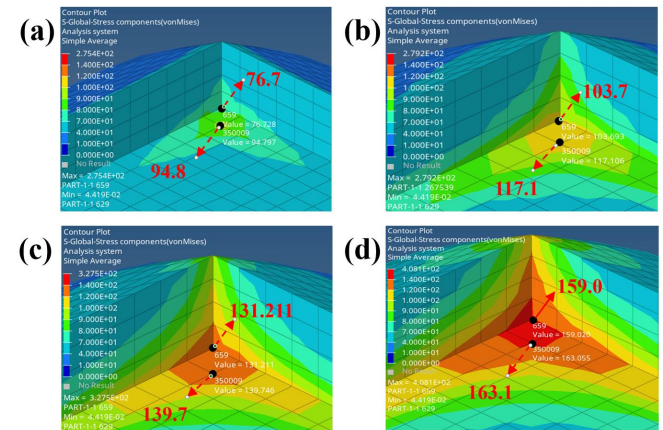


Fig. 3. Von Mises stress (a) without lid; (b) when the CTE of lid is 10 ppm/°C; (c) when the CTE of lid is 14 ppm/°C; (d) when the CTE of lid is 18 ppm/°C.

TABLE II  
MATERIAL PROPERTIES

| Material   | CTE (10 <sup>-6</sup> /°C) | Young's Modulus (MPa) | Poisson's Ratio | Density (10 <sup>3</sup> Kg/m <sup>3</sup> ) |
|------------|----------------------------|-----------------------|-----------------|----------------------------------------------|
| Lid        | 14                         | 188000                | 0.24            | 7.7                                          |
| Lid attach | 300                        | 600                   | 0.35            | 1.1                                          |
| Die        | 2.8                        | 131000                | 0.28            | 2.3                                          |
| TIM        | 117                        | 2.1                   | 0.4             | 2.7                                          |
| UF         | 29.5                       | 9000                  | 0.35            | 1.52                                         |
| Solder     | 24                         | 39000                 | 0.3             | 7.38                                         |
| Substrate  | 14                         | 24500                 | 0.22            | 1.9                                          |

### B. The Influence of Lid on Stress

Since the lid obviously affects the deformation of package, we are also very concerned about the influence on the stress of package. The stress of UF is studied as a representative, which could be used to evaluate interface delamination.

Firstly, we extracted the Von Mises stress of UF at the edge point of the chip and found nearly a linear trend with the increasing of CTE of lid for different forms of stress, as shown in Fig. 3.

Von Mises stress is an equivalent stress based on shear strain energy. There are various other types of stresses, such as maximum principle stress (P1), minimum principle stress (P3), max shear stress and so on. Here, we summarize the relationship of Von Mises, P3 and max shear stress with CTE of lid, as shown in Fig. 4.

For CTE of lid of 14 ppm/°C, it is in a critical state, we are concerned whether a slight change near this position will cause additional stress. So, a few more points are added around 14 ppm/°C for CTE of lid, as also shown in Fig. 4. It is clear that all types of stress of UF increases monolithically with CTE of lid, indicating that the mechanism for UF stress is basically similar. Although different modes exist in the variation of package warpage, the stress of UF area has little effect, which may be attributed to relative larger distance between UF region and the edge of package. Finally, it should be pointed out that the stress level of the UF will be significantly underestimated in lidless structure, larger CTE of lid will lead to larger UF interface stress. Only Von Mises stress is analyzed subsequently as a representative.

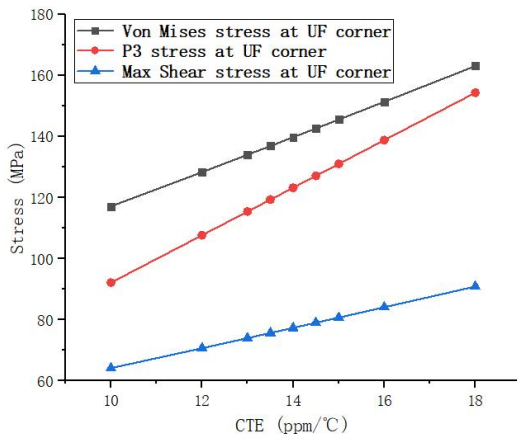


Fig. 4. Different forms of stress at UF corner.

### C. Other Influence Factors

The above analysis basically briefly explains the main influence of lid on package warpage and stress. But there are still other factors worth discussing and analyzing.

#### 1) Viscoelasticity of UF

It is well known that polymers are generally viscoelastic, and viscoelasticity affects the value of UF stress. Here, we also analyzed the change of UF stress with different CTE of lid when using viscoelastic properties for UF [7], as shown in Fig. 5. It can be seen that when applying the viscoelastic

parameters, the stress of UF will be lower than that of linear elasticity due to the consideration of stress relaxation. However, the overall trend has not changed, i.e., the UF stress increase with the increase CTE of lid. But the relationship of stress with temperature will not be linear anymore.

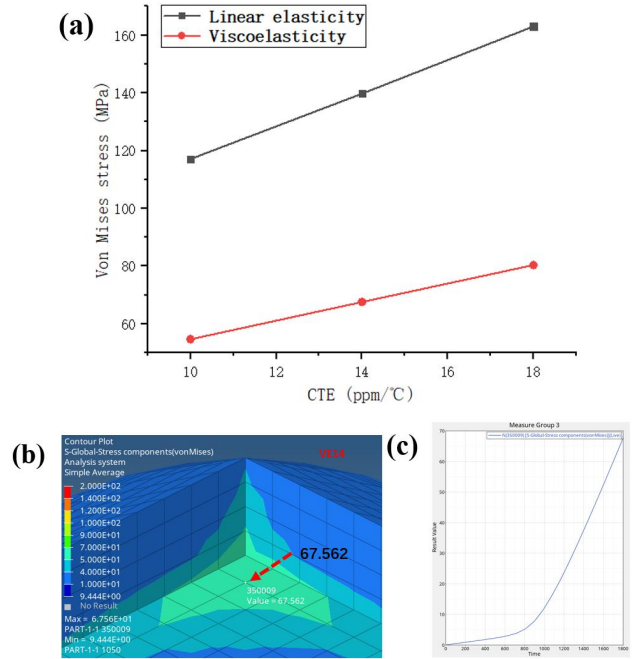
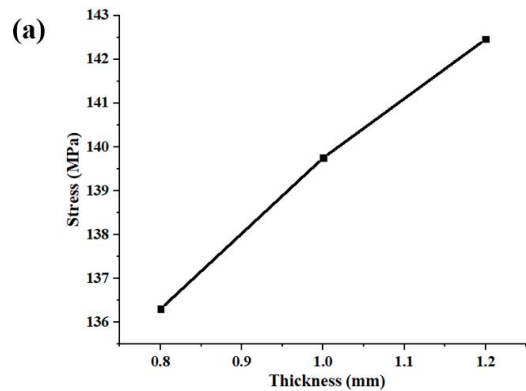


Fig. 5. (a) Von Mises Stress comparison between linear elasticity and viscoelasticity; (b) Von Mises stress of UF with viscoelasticity; (c) Von Mises stress change with time with viscoelastic properties of UF.

#### 2) Modulus and Thickness of Lid

In addition to CTE of lid, the modulus and thickness of lid may also have an influence on UF stress. This is mainly affected by the degree of constraint to the substrate. It can be predicted that when the lid modulus increases or the lid thickness increases, the confinement effect also increases, which will lead to a decrease in the overall deformation of package but an increase in stress of UF. Next, it will be verified by FEA.



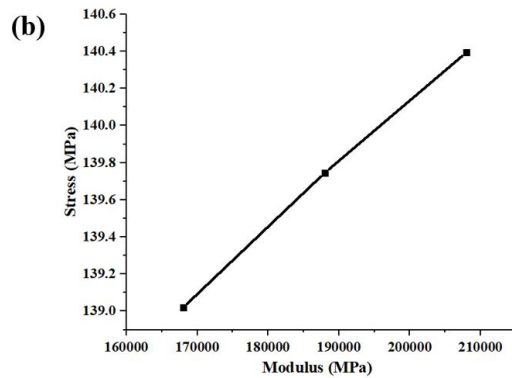


Fig. 6. Effect of lid on UF stress when (a) lid has different thickness; (b) lid has different modulus.

The effect of thickness and modulus of lid on UF stress are shown in Fig. 6. It is clear that our prediction is correct: the lid modulus increases or the thickness increases will cause an increase in stress of UF.

#### IV. CONCLUSION

- 1) Lid significantly affect the deformation or warpage in package. For lid with different CTE, three different warpage modes are presented.
- 2) However, for stress of UF, larger lid CTE will lead to larger UF interface stress, regardless of the warpage in package.
- 3) The stress level of the UF-related interface will be significantly underestimated in lidless structure.

#### V. ACKNOWLEDGEMENT

The work was supported by the National Key R&D Program of China (no. 2020YFB0408700), the National Natural Science Foundation of China (no. 52003289), Shenzhen Basic Research project (no. JCYJ20180302145742105, JCYJ20210324115611030), the Youth Innovation Promotion Association CAS (no. 2021363), the SIAT Innovation Program for Excellent Young Researchers (no. 201803), SIAT CAS-CUHK Joint Laboratory of Materials and Devices for High Density Electronic Packaging, and the Guangdong Provincial Key Laboratory of High Density Electronic Packaging Key Materials (no. 2014B030301014).

#### REFERENCES

- [1] Zhang Z, Wong C P. Recent advances in flip-chip underfill: materials, process, and reliability[J]. IEEE transactions on advanced packaging, 2004, 27(3): 515-524.
- [2] Zhang Z, Wong C P. Flip-chip underfill: materials, process and reliability[J]. Materials for advanced packaging, 2009: 307-337.
- [3] S. Shao, Y.L. Niu, J Wang, R.Y. Liu, S.B.Park, H.Y.Lee, G.R.Ahmed, L. Yip, Comprehensive Study on 2.5D Package Design for Board-Level Reliability in Thermal Cycling and Power Cycling, ECTC, 2018, 1668-1675.
- [4] V. Pham, J.F. Xu, K. Pan, J. Wang, S.B. Park, C. Singh, H.Y. Wang, Investigation of underfilling BGAs packages-Thermal fatigue, ECTC, 2020, 2252-2258.
- [5] M.C. Paquet et al, Underfill Delamination to Chip Sidewall in Advanced Flip Chip Packages, ECTC, 2009.
- [6] T. Sinha, et al, Geometric Optimization and Mechanical Risk Mitigation in 2.5D Flip-chip Packages using Parametric Finite Element Analysis (FEA) Simulations, Itherm, 2016.
- [7] C. Zhong, C. L. Li, L. Lu, et al., "Characterization and Verification of Viscoelastic Constitutive Parameters of Underfill Material," International Conference on Electronic Packaging Technology (ICEPT) 2021.